Go-Live Bill of Materials and Deployment Readiness

REVA AICTE IDEA Lab Portal on AWS with everything real: Microsoft sign-in, live Razorpay payments, bookings, email and backups

Ref: IDEALAB-BOM-2026-03
Date: 29 September 2026
Currency: INR

Running cost per month (core)	Per month with extra security	First month to go live (core + Claude Pro)	First year (core + Claude Pro)
₹2,378	**₹4,168**	**₹4,878**	**₹31,036**

Status: the software is complete and tested (74 automated tests, security audit, browser tests, Docker build all passing). What remains is accounts, keys and approvals, listed in section 4; no more coding is needed to go live.

1. Costs

No.	Item	What it is for	Type	Rate	Monthly	Year 1
A. AWS hosting (required)						
1	**AWS server package**	EC2 t3.small server, 30 GB SSD, static IP, S3 + snapshot backups, SES email, data transfer (details: IDEALAB-BOM-2026-01, section A)	Monthly	incl. 18% GST	₹2,378	₹28,536
Total A					**₹2,378**	**₹28,536**
B. Extra security on AWS (recommended)						
2	**Security package**	Web firewall (WAF), CloudFront + DDoS protection, GuardDuty, Inspector, Secrets Manager, CloudWatch alarms (details: IDEALAB-BOM-2026-01, section B)	Monthly	incl. 18% GST	₹1,790	₹21,480
Total B					**₹1,790**	**₹21,480**
C. One-time						
3	**Claude Pro (1 month)**	AI assistant for the go-live setup and any fixes	One-time	₹2,500 / month	-	₹2,500
4	**Razorpay live account + KYC**	Activation for real payments (business documents, bank account)	One-time	Free	-	₹0
5	**Microsoft sign-in registration**	App registration in REVA's Microsoft 365 by REVA IT	One-time	Free	-	₹0
6	**Domain + HTTPS certificate**	REVA subdomain (e.g. idealab.reva.edu.in) and Let's Encrypt certificate	One-time	Free	-	₹0
Total C					**-**	**₹2,500**

No.	Item	What it is for	Type	Rate	Monthly	Year 1
D. Usage-based (paid only when students pay; not in totals)						
7	**Razorpay live payments**	UPI, Indian cards, net banking: 2% + 18% GST on the fee; international/Amex: 3% + GST	Per payment	2% + GST	see example	-
Example: ₹20,000 of 3D-print payments in a month → about ₹472 in fees (can be passed to students as a convenience fee).						
YEAR 1, CORE: A × 12 + C					**₹2,378**	**₹31,036**
Year 1, with extra security: (A + B) × 12 + C					**₹4,168**	**₹52,516**

AWS amounts are estimates for the Mumbai region at an assumed 1 USD = ₹85, including 18% GST; confirm with the AWS Pricing Calculator. A 1-year EC2 Savings Plan cuts the server cost by about 30–40%.

2. Payments: what students pay, fees and GST

2.1 What students pay for (as the portal works today)

Service	How the price is set	How it is paid	Razorpay fee?
3D printing	Calculated on the server from the uploaded model: weight (g) × material price per gram × quality (Standard ×1, Medium ×1.2, High ×1.5). Minimum ₹5. Lab staff set the price per gram per material in the dashboard.	Online: Razorpay (UPI, cards, net banking). Unpaid orders auto-cancel after 30 minutes.	**Yes**
PCB manufacturing	Calculated on the server from board size, layers, quantity and finish.	Offline (cash/UPI at the lab); staff mark the order Paid.	No
Components (indents)	Sum of the unit costs set in Inventory × quantity; free items cost ₹0.	Offline; staff mark Paid.	No
Events, slot booking	Free	-	No

2.2 Example prices per student (from the portal's own formula; demo material prices)

Example 3D print (20% infill)	Material	Quality	Weight	Student pays	Razorpay fee (2%)	GST on fee (18%)	Lab receives
Small part / keychain (5 cm³)	PLA ₹5/g	Standard	3 g	**₹14**	₹0.28	₹0.05	₹13.67
Phone stand / enclosure (20 cm³)	PLA ₹5/g	Standard	11 g	**₹55**	₹1.10	₹0.20	₹53.70
Phone stand / enclosure (20 cm³)	PETG ₹6/g	High	11 g	**₹101**	₹2.02	₹0.36	₹98.62
Large model (100 cm³)	PLA ₹5/g	Standard	55 g	**₹273**	₹5.46	₹0.98	₹266.56
Large model (100 cm³)	PETG ₹6/g	High	56 g	**₹503**	₹10.06	₹1.81	₹491.13

PCB examples (paid offline, no Razorpay fee): 50×50 mm, 2 layers, 5 boards = ₹2,050 · 100×100 mm, 2 layers, 5 boards = ₹2,200 · 100×100 mm, 4 layers, 10 boards, ENIG = ₹7,200.

2.3 Monthly and yearly: money collected and payment fees (usage assumptions)

Online 3D-print payments	Orders / month	Average order	Collected / month	Razorpay fee + GST / month	Lab receives / month	Collected / year	Fees + GST / year	Lab receives / year
Low	30	₹120	₹3,600	₹85	₹3,515	₹43,200	₹1,020	₹42,180
Medium (planning figure)	100	₹150	₹15,000	₹354	₹14,646	₹180,000	₹4,248	₹175,752
High	250	₹200	₹50,000	₹1,180	₹48,820	₹600,000	₹14,160	₹585,840

Fee = 2% of each payment (UPI, RuPay, Indian cards, net banking) + 18% GST on that fee. Foreign cards and Amex: 3% + GST. The order counts are assumptions for planning; the Sign-in Log and order history will show real numbers after launch.

2.4 GST summary

GST item	Rate	Who pays	Where it appears	Included in this BoM?
GST on AWS bills (hosting, security)	18%	The lab (to AWS India)	Monthly AWS invoice	Yes, in A and B
GST on Razorpay fees	18% of the fee	The lab (deducted from each payment)	Razorpay invoice; deducted before settlement	Yes, in 2.2 and 2.3
GST on Claude Pro	As charged	The lab	Anthropic receipt	Assumed inside ₹2,500
GST on what students pay (3D print, PCB, components)	Decision for REVA accounts	Student, if applied	Not added by the portal today; prices shown are the full amount	No (see note)

Whether GST must be charged to students (and whether GST paid on AWS/Razorpay can be claimed back as input tax credit) depends on REVA's GST registration and how the lab's charges are classified. Confirm with REVA's accounts team. If GST must be shown on student bills, the portal can add it as a separate line on the price and receipt (a small code change).

2.5 Overall month and year, including payments

Overall (medium usage)	Per month	Year 1
AWS hosting (A)	₹2,378	₹28,536
Razorpay fees + GST (medium usage)	₹354	₹4,248
Claude Pro (one-time)	-	₹2,500
TOTAL COST, core	**₹2,732**	**₹35,284**
Total cost with extra security (+ B)	**₹4,522**	**₹56,764**
Money collected from students for 3D prints (medium usage)	₹15,000	₹180,000

Refunds: made from the Razorpay dashboard; Razorpay normally keeps the original fee (check current terms). Settlement: money reaches the lab's bank account about 2 working days after each payment. Fees can be passed on to students as a convenience fee if REVA prefers.

3. What is ready (software)

Part of the portal	Status
Student sign-in with REVA Microsoft accounts only; staff roles; emergency admin	**Done**
Sign-in log with counts and CSV (separate, append-only database)	**Done**
Events and registrations (safe seat counting)	**Done**
Lab slot booking with approve/reject, lab closures	**Done**
Components / indents with server-side pricing	**Done**
3D printing: STL upload, server-side price, Razorpay payment + webhook, auto-cancel unpaid	**Done**
PCB orders, inventory, gallery, projects, CMS, PDF reviews	**Done**
Security: role checks, injection protection, upload checks, rate limits, security headers	**Done**
Docker Compose (app + password-protected MongoDB), nginx HTTPS config, nightly backups	**Done**
Supabase → MongoDB data migration script	**Done**
Automated tests (74), CI on GitHub, deployment/testing/security documentation	**Done**

4. What is left before launch (accounts, keys, approvals)

Step before launch	Who	Time	Status
AWS account, EC2 server, IAM role for email, static IP, security group	Lab / you	~1 hour	**To do**
Point the domain (idealab.reva.edu.in) to the server's IP	REVA IT	Same day	**To do**
Register the portal in Microsoft Entra ID; give MS_TENANT_ID, MS_CLIENT_ID, MS_CLIENT_SECRET	REVA IT	~15 minutes	**To do**
Razorpay: complete KYC, switch to Live, generate live keys, add webhook	Lab (accounts)	2–5 working days (approval)	**To do**
Amazon SES: verify sender domain, request production access	Lab / you	~1 day (approval)	**To do**
Fill .env on the server, docker compose up, create emergency admin, nginx + HTTPS	You	~1 hour	**To do**
Run the Supabase data migration; check users, orders and files came across	You	~30 minutes	**To do**
Delete old Supabase/Firebase/Google keys; new JWT_SECRET; remove .env from git	You	~30 minutes	**To do**
Go-live test: real sign-in, ₹1 live payment and refund, event registration, slot booking, backup restore	You	~1 hour	**To do**
Security package (WAF, CloudFront, GuardDuty, etc.)	You (with Claude)	~half a day	**Optional**

Longest waits are Razorpay KYC and SES production access; start those first. Realistic time to launch: about 1 week, mostly waiting for approvals.

5. Production settings (.env on the server)

Setting	Where it comes from
`PUBLIC_URL`	https://idealab.reva.edu.in (your domain)
`JWT_SECRET`	New random 64 characters (openssl rand -hex 32); never reuse the old one
`MONGO_ROOT_PASSWORD, MONGO_APP_PASSWORD`	New strong passwords (letters and numbers)
`MS_TENANT_ID, MS_CLIENT_ID, MS_CLIENT_SECRET`	REVA IT, from the Entra ID app registration (redirect: <PUBLIC_URL>/api/auth/microsoft/callback)
`RAZORPAY_KEY_ID, RAZORPAY_KEY_SECRET`	Razorpay dashboard, Live Mode → API Keys (rzp_live_...)
`RAZORPAY_WEBHOOK_SECRET`	Razorpay → Webhooks → <PUBLIC_URL>/api/payments/razorpay/webhook, events payment.captured and order.paid
`EMAIL_FROM, AWS_REGION`	SES-verified sender, e.g. IDEA Lab <no-reply@idealab.reva.edu.in>; ap-south-1
`BACKUP_S3_URI (optional)`	s3://<your-bucket>/idealab for off-server backups

6. Launch day, in order

1. On the EC2 server: install Docker, git, nginx, certbot; clone the repository (DEPLOYMENT.md section 4).
2. Create .env from .env.example with the values in section 5; chmod 600 .env.
3. docker compose up -d --build; check docker compose ps shows both services healthy.
4. docker compose exec app node build/scripts/create-admin.js --email you@reva.edu.in --name "Your Name".
5. nginx + free HTTPS certificate: certbot --nginx -d idealab.reva.edu.in.
6. Migrate the old data: node build/scripts/migrate-from-supabase.js --dry-run, then without --dry-run.
7. Add staff (Dashboard → User Management) and switch the domain on for everyone.
8. Nightly backups: add deploy/backup.sh to cron; test one restore.

Full commands: DEPLOYMENT.md in the repository. Testing: TESTING.md. Security review: SECURITY.md.

Prepared by	Reviewed by	Approved by
Name, signature and date	Name, signature and date	Name, signature and date